

Adam Safron ~ Active Inference Insights 006 ~ Psychedelics, Emotions, The Body

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welcome back everybody to the sixth episode of active inference insights I am back today with another fantastic guest Adam saffron Adam is a postdoctoral research fellow at John's Hopkins Center for psychedelic and Consciousness research with deep roots in the active influence framework Adam's work has explored the nature of preferences and motivation mindfulness Consciousness agency and psychedelics in addition to this he might well be our only guest thus far who has written a paper titled what is orgasm and they lifted 280 kilos for multiple reps uh in sandals my I add uh note by the way that these two domains need not be related Adam thank you so much for joining me and uh welcome to the show so good and I'm really glad you could join us and as I said I mean I'm always Blown Away by the diversity of interest and and research topics that our guests bring to the table so it'll be great because I think we we're going to have a wealth of things to cover the first thing I wanted to ask is about the body um you write in um one of your papers bodies represent near ideal initial systems for Learning and inference with this prototypical object and causal system providing bases for further modeling I wanted to ask because I've been interested in what yakob how called The evidential Boundary where can we kind of circle off the internal and the external uh with respect to mark of blankets and act of inference so I wanted to get your take on whether you thought that what do you think of the body is it a prior that we could say the mind the cognitive system uh it's incorporating uh in its in its sort of uh predictive schema or is it itself an objective inference that the mind um you know conducts probabilistic reasoning over yes to which one on so I was s in that particular paper um I'm trying to provide a count of basically how core knowledge gets uh loaded into Minds so you know like we need something like core knowledge but what's a biologically plausible account of how we might arrive at it and so then the idea is you're basically using um the body as um basically a way of structuring structuring um learning curricula to make them tractable um for a computationally bound system and a um with limited learning opportunities like making the most with the least and the idea is you

create basically the body acts as a near um ideal like training wheel system to figure out where the conditions for inference are fruitfully constrained and then you can use that in the structural learning process sure and so um you can think of it as so the body would be both um in way an evolutionary prior and that the specific ways in which it's some uh you know like for instance like the um R Fifer I think speaks really well to this in terms of like the way that the the The Limited shoulder the the range of what the mobility of the shoulder Droid for instance already would would make your your arm go out in front of you and then like be more likely to grab things and where they're like well look right there at the right distance for me to both manipulate them and to observe them and so those would be the kind of like evolutionary priorities you might expect but it's like a in the bootstrapping process it would be um you know you could also think of it though as um like core knowledge to get to there it would be may maybe not necessarily an evolutionary prior but a very reliably learnable posterior so or an empirical prior um so you know I I think depending on like but it might be so reliably learned and so canalized uh eight ways from Sunday that it might as well be an evolutionary prior even though there is like this mediation by the learning that takes place but it's basically in the bag because you did it because nature did it so well um so whether you would call it a prior or posterior um depends on your perspective and and um some other people who've um kind of talked to the body is a prior like uh Micah Allen had a good paper uh the the body is prior talking about the way in which for instance um our inceptive states as particularly stubborn predictions would help to make sure that like the whole like generative model is basically uh orbiting around allostasis and so like so yeah yeah does it make does it make intellectual sense to you to carve a dividing line uh like an evidential boundary or can we is it is it more sort of elegant to just say it's philosophically vague so yeah so um so I guess like in that tradition of like the so like the body is first prior um there's also like to work with like um anach and it and um others where it's like us starting like mentalizing homeostasis where it's like you you're um part of this learning curriculum is uh multi-agent and uh coupled like like we're starting within someone else's bodies and then even when we're on the outside we might as well be because we're so dependent and so like uh and within like this also this um um Ral um feran context or this roboticist um and the overlap with like Rodney Brooks also um similar kind of thinking with the world is its own best model and like emergences and more and and a lot of the lifting being done by morphological intelligence um you know even when you know you're on the outside of the womb uh you're uh intensely coupling with others in an extended mind sense so it's like uh it seems like you definitely uh you can't just say take

it like at the body boundary then you the Mind stops you know it's um yeah this account this is a really interesting Avenue I've been thinking about um Clark and charma's 98 extended mind hypothesis paper in the lens of active inference and I've been sort of informed in my thinking by two people um I've really mentioned one of them on the earlier podcast which is Manuel delander and his most recent book um M uh what's it called material phenomenology he gives us all uh mechanistic account of vision and he critiques the an activist tradition on this count that basically what they want to call an intrinsic relation I.E the coupling itself is constitutive of the identity so in the coupling between agent and Arena you actually form the definition of agan Arena he criticized it that that what actually that really is is a causal extrinsic relationship so that's one strand and then the other is Thomas metzinger and Thomas a did a podcast with professor friston and he spoke about how the notion of inaction might be quite misleading because it kind of implies that there is there might be a subtle reoc of a cartisian ego where you have a enacting b so I want you know I want to kind of pick your brain on that because it's interesting you B up extended cognition and the fact that the Mind might its identity might go beyond the the remits of the body are we are we at in danger there invoking what danda calls an intrinsic Rel relation where really what we're looking at is just causal connections uh and we're not we shouldn't actually be mistaking that for identity relations I'm not sure um and and you know I might also um like in that paper I imagine I like kind of antagonize more than a few people and actually of like going uh neoc cartisian in a way like suggesting like something like quasi homuncular and thees is like as like a ways of like understanding like agents and their relations to the world in terms of like the way they're situated and that um internal having an internalized representation uh pegged according to this uh reference frame of an observer um from an ego Centric perspective uh would do a lot of like heavy lifting in the process of structure learning and making things tractable um so um the yes it's like I guess yeah you might have to like loop around a few times you know in terms getting me to dock properly but uh I guess the way I'm thinking it's like like mind as model has to be like extended and it's like the um the ways in which like the affordance landscape is so dependent on these couplings to to like to to go to internalist is in a way miss the point about what lets you do anything what lets your natur but I I do have sympathies for the internalist account in terms of making room for things like um representation so that you can do things like um offline learning and imaginative planning and like you know basically like the world is its own best model whenever possible it's morphological compute a compute you know uh emergent com um Mor uh you know from the couplings of the organism environment you put don't just like push your

predictions down to to the lowest level of like the brain push them out into the environment
yeah yeah yeah yeah yeah well EXA I mean yes that that I guess was the the question I wanted
to come to which is um so it's all very nicely interwoven you're right you invoke this notion of
the kind of Quasi hular systems um and you mentioned the frame problem um which is a
really interesting problem to me which is kind of how can we be uh intelligent in an ignorant
manner uh how can we zero in on personal information in the world given the preponderance
of sense data intelligent in an ignorant manner I love that um thanks it's it's it's uh it's not mine
I'll shout out John vvi I mean he's the one who's kind of outlined the frame problem to me
more than more than anyone else you so I this is kind of a leading question because I have a
sense of where you're going to how you're going to answer it given what you just said but can
the frame problem be solved by just kind of a single conscious homunculus at the tip of the
prefrontal cortex and if not how deep do these or how deep do you imagine that these quasi
homuncular or non all these mindless cognitive agents as manand would call them how deep
do they go towards the skin I'm sorry I don't I've gone right into the deep end on these but why
not no no it's good um so it's seems the extended and active nature of Minds um helps to
curtail like frame problems and that you're you're always like in the world in particular ways
and if like you're coupling with the environment is creating basically like is basically like in a
way creating like an enslaving Center manifold for the whole system that you you you always
have to keep coming back to to stay on top of this action not just to stay on top of this action
perception cycle and on top of um this inter interchange um you know the full scope of
relevance as if you're always kind of like being brought back home and grounded in this
common task of cyber control that that seems to like take some of like the I I don't if it quite
defangs the frame problem but it helps to curtail it a bit um the and so then you know but in
terms of also like taking the the the range of um you know potential relevant affordances it's
still you know a big space um but still you know the idea of like orbiting it around um an
agent's model of itself and interaction with the world um I don't know how far um homuncular
for instance representation gets you with respect to that but in a way it um the idea of like
basically one of the core things that you're modeling is the living body the idea at the center of
your experience this um uh you know so so the hun I guess so guess unpack hun so like the the
thing that people are worried about would be right a homuncular regress where just like there's
a little man in the head that's doing the observing and you've explained things and so the
reason I kind of um start to push on this issue of like trying to resurrect homunculi and things
like cartisian theaters is the idea is that the the folk intuition and and like is coupled to the

phenomenology in a way that actually might reflect the data structures might actually reflect the way the system is organized to reflect the principles of the controller right and and so you know the importance of taking your perception and giving it a coherent perspectival reference frame so something you know theater esque in terms of you know having like the right geometric relations of the things of the world their relationship to you but then also for instance though the idea of like an observer that helps that watches the screen in the sense of your embodiment as a central data structure being in close communication with uh basically the systems by which you would do a visual perceptual synthesis because you know you're you know what your body is doing will inform what you're likely to be what it means what's coming in on your retina and or the visual spatial system the somata spatial should tell you about the visual spatial and vice versa and it's interesting that in robotics these are actually the primary data structures they so like in the collaborations I've done with like Tim Ver Bale and and Oren Etzioni like in their latent slam systems one of the primary things for situating the robot in the world is basically a conjunction of views and poses it's like and and they're almost like one data structure and that it's you know one of the best you know where I'm looking you know where I'm where my sensors are oriented are going to be highly informative for what I'm looking at and also just where my the way my body is configured I don't think this quite gets in this isn't quite brought into their systems but you know the range of relevant importances what am I like be seeing well your the your uh what your body is doing will be highly relevant to that also and so the mutual information should be very strong closely coupled right yeah it it it kind of brings me back to the to the first question I guess which is I mean I could probably reword that now using the terms you just said I mean I think internal and external an act of inference can be considered a kind of relative misnomer because if you know I like Maxwell's rewording of it as uh tracking and tracked or modeling and modeled I think that's quite an effective way of resolving the focus here so I guess the way of rewording that question is you know are we invoking a homunculus and the C in theater are we a risk therefore of kind of thinking that there's a genuine ontological homunculus which is conducting variational Bayes and inference uh and and has a model over which it does computations I think like that rubs a lot of people the wrong way because for a lot of people it's just dynamical systems and so you know do these kinds of questions actually call into question the more fundamental dichotomy of tracking and tracked and distinct ontologies distinct entities in the world potentially and the so you know maybe I'm asking for too much but I want to have it all I want to have um I

an active couplings in the streets and like reations and at Sheets you know like um you know to the extent possible like you know there's even need to appeal to like some sort of model representation for much of what I'm describing of this like Mutual contextualization of like these different sensory systems in terms of they're already coordinated just by having a certain situation on the body itself and in the world and the way like the um the the information is cascading in the context of the inactive coupling and so there there's a sense you know the the the model of the agent is extended into the world its mind is extended already there and a lot of like the the work I'm talking about we like the homonculus is the body itself there's no need for like an in addition representation of it um I do though suggest we might have though um you know such basically a uh a data structure you would say or um and and maybe even um potentially involving uh you know I I I put this forward basically as a kind of pred that like nothing rides on in the sense of like uh if I'm wrong sure uh but if I'm right could be interesting so but the specific idea is that there might be sub networks of the brain that actually um have a geometric organization about them in terms of you might you would even have like within cortical Hub structures potentially a um geometric correspondences through the manifold activity in those networks and what's being represented so you you could even get like homuncular in terms of like the network is almost little personlike on some portions and the reason you might want to do this is that um within geometric deep learning like if you give a network the same geometry essentially um not like a physical Network in terms like well guess it would be there in the connect eventually but like it could be just like a dynamic emerging pattern um effective connectivity but like if the the network though has the same structure as that which which it is um representing or modeling or trying to track you end up getting potentially orders of magnitude greater efficiency yeah that's really cool that's really so it's it's possible that like there could be in addition to like most of what we're doing is like effortless Mastery and the vular story you never need representation you never need internalized model and most of it's and you would want that too you know that's right that's quicker more efficient but you might also on top of that for the sake of things like imaginative planning things like having more nuanced control of the sensory motor engagement you might also have representations of this um and um potentially even uh uh shockingly cartisian accounts like in terms of Highly centralized and even kind of looking uh with a GE ometry that would that would be uh of a similar shape to the body itself yeah that that's fascinating I mean these uh these strange Loops as hofstead would call them are are seemingly never ending the body having Loops of itself the body having Loops of the world I mean it comes

back to I think um kind of the cybernetic good regulator theorem and and like is that saying you know there's this kind of confusion is that saying that all good Regulators have a model of the world or are a model of the world and I think I think I don't know I mean I don't know whether you think that's still a Live question in active inference I mean to be honest to be completely candid and yes I'm a relative Layman with a Year's experience of thinking about active inference and not understanding the maths and the physics but that's okay I think there's still a kind of question there which is and Cole to be honest was quite neutral on on it when I asked him which is like what is it what what will are internal States um yes internal so so it makes sense that internal states have or embody have embodied beliefs about the external States I mean that is at the heart of like a likelihood mapping so fine but is that it uh is that what we're going to give them is these beliefs and what is it to embody a subpersonal belief is it just an instantiate like I think people find it very difficult to basically move away from the physical mechanics to the statistical mechanics and I think there needs to be a bridging mechanism so I don't know if you had any thoughts just on like the broader way that a sort of layman audience including myself could make those intuitive or those non-intuitive distances between statistics and physics or the physical Material World a bit more intuitive like we're working with this account of like you know blankets of blankets of blankets and um I guess the way I'm interpreting is like a internal internal States course graining external States and then Nest within there internal States course graining external States so on and so forth like a an a hierarchy or a hierarchy of nested nested blanketed structures um you know giving you the these multiple levels of resolution that you can unpack as needed and so I guess the the thing I'm wondering is whether like uh I like I wonder whether a uh all so so it seems like you know none of it works without like the inactivist principles like without like the um the the the structuring of what's Happening by intelligent morphologies or um or without just like that um uh the nature of the coupling of the system in this world providing like the grounding and the semantics um and the interest none of it gets off the ground but but the story still it can be told just just at the level of like the system and its extended phenotype as this one single model and like where do you draw this boundary what's internal what's external it's like it depends on the scale you're thinking about it right um and so I guess the place where I wonder is where going to need internalism is when we get to Consciousness yes basically like so so like so I guess you might even call me like a kind of like pan psychist of a stripe and psyche in terms of understood as like mind but then when you get to like basically Consciousness in terms of something like uh a system for we might say there is something that it feels like to be that

system then and for that can do things like um imagine for instance in dream like for for that and those elements of experience it seems like we're going to have to at some point invoke um enough uh a a a a we're going to need a blanket uh uh a a set of attractors that are internal and can uh basically Break Free of the immediate engagement with the world that's part of our experience and so like for people I think to actually like account for phenomenology to account for like you know like a the things that that matter to us you know eventually getting to you know things like agency even like free will we're going to need to at some point um go internalist but like in that process though you know it's just but uh still the story for most of evolution uh you know but still like if you go in that direction though and you lose sight of the the inactivist principles well then you you basically find yourself in the same places that I guess you could say like the mainstream uh machine learning goes astray sometimes you think you're going to handle the problem just with scale or raw compute or or you don't understand like the kinds of uh learning curricula you would need to bootstrap intelligence systems or how they would need to be deployed to to you know achieve goals in the world um so uh yeah so again like I I think I end up like occupying a very awkward position that's like you know the um radical and activists will be you know you don't need to invoke these things like um uh uh cogn this might be like you know I know you're leaning way too hard and saying that like embodiment is necessary and and you're just and but I think that is the case though like it's the um I think everyone was kind of right in just like emphasizing account of what's going on yeah yeah yeah I mean I I I haven't St my my personal claim but I don't disagree and I think that Nuance is is always going to be right in some fundamental sense um all theories are wrong they're just wrong at different you know to different degrees all metaphors are wrong some metaphors are useful right exactly I mean I I was Wonder you know it's funny that it's not funny it's interesting that you say that the an activist principles are necessary um it'd be great to have like a pure inter or I can't label people but like someone like yob Howie who you know if he was here might again I'm not putting words in his mouth so it be curious to see what he thinks might ask whether it's uh conceivable that a kind of compute only system could do variational Basi and inference um given its priors I mean I think I think another useful distinction actually to to unpack for our audience and something that I found really really useful is this distinction between uh having a sort of epistemically viable model model of the world so you are actually you know it's descriptive it it accounts for the sensations that it's uh observing or it sensation it's receiving and a normatively viable model of the world I.E one where you aren't just dispersing into the heat B but you can maintain your own sort of

autopoietic organization and I'm wondering whether you think it's conceivable to any degree whether both sides of that equation could be fulfilled by a kind of sheltered off super intelligent homunculus sheltered off and it never comes into contact well I mean so so it it comes into contact uh vicariously through the markup blanket so so there I'll permit active and sensory States um but again vicariously it has to do this kind of variational basin inference um based on its its model and we can kind of get into the maths of that a little bit of like where that model comes from in the first place might not be quite what I think we're we're driving at but like one thing I would wonder in terms of like um like saying like no you don't need um um you know things like uh active learning to be like intell to you don't need to like invoke things like extended mind Concepts you don't need to invoke things like Niche construction like in theory like you could do everything just from uh a smart enough basing Observer like I guess would that be like the complete class Ser is like like like uh well yes like almost like has to be the case but that that in theory you could do it that the the an activist um elements that being appealed to like if you're going like in theory if you want to like build like a like an AGI or something uh you wouldn't need to but then I guess the question is like if that in theory though like in theory like a three layer neural network can compute anything right question it just might have to be so big it doesn't fit in like the universe and so it's like if that in theory can't become in practice given like available compute available to systems we would real we would feasibly be able to create then it becomes I think a little bit moot although still interesting about like that case of like uh like is it a matter of in principle or in practice so um yeah it's where philosophy has its limits perhaps um where conceivability arguments have their limits you know a question that popped up into my mind is boot strapping up uh intelligent uh or you know um kind of inv vitro intelligence um I think I think one question I've always had is the priors that um are encoded in the generative model kind of make sense in this normatively viable sense so people just think of homeostasis we have these kinds of um yeah prior expectations about the states that we expect to be in and will act accordingly to to confirm that model I think people might have a bit of a difficulty applying the same logic to um perceptual inference so perceptual inference uh you know this idea that basically you're inverting Bay you are trying to derive the state from the observation I guess the the question that I have is if that's a sort of $P(x \text{ given } y)$ your your probability of the state given the observation obviously that's done through some form of the variational base people might be asking well where does the p of X comes from where does your prior of States before observations um I think what might be the denter in pom DPS where does that come from like

is that encoded in the phenotype in the same way that homeo homeostatic uh imperatives might be or yeah or is it the the D Matrix or the like you know I think it's the yeah I think it's the D Matrix again I'm not a computation by your specific initialization within the environment and like getting the ball rolling the yes exactly exactly it's you know it so it's like it seems like it's like we know like without the inductive biases like none of it works the structural learning is just impossible you know with without the evolutionary priors without like some sort of like you know intelligent initialization I guess the place where um I tend to have a I guess like a heterodox View and that like it's like well I grant like we need something like core knowledge we need basically prior to to lean on um I tend to think that um EV like natural selection had fat fingers when it came to like like doing the design like both in terms of like potential complexity mismatches between inherited information and then like what comes afterwards but also due to like in like not just like in uh why you know people sometimes refer to like the difference and complexity between like the genome and the connectome for instance you know but obviously you know the genome you the epome and there's a lot more inheritance than that like just in like the cell like a lot there's a lot of inheritable elements that that go off so the question is like but even then there does seem to be some difference in like you've squeezed through the bottleneck of like an ovum and so from this one cell you reconstituted the rest and this kind of like kind of like Auto encoding like you get to pass through a bottleneck here and somehow but from this you're now though going to con reconstitute a a network of effective of functional relations and effective connections like a full like brains connectum and is you know is there enough in there to likely things the way you might need to um the way I tend to like kind of think about it is like so one heuristic is um you know the more so like what um kind of knowledge might might we need what might be the nature of the representation or just the instantiation AI then the more complex it is the less likely it is that that is in a more direct sense uh specified by so it's like yeah you know if we're talking about like you know um so for instance like you can think of like uh there's like a prior encoded for instance in like the the Frog brain of like the superior colliculus of like potentially of um based on sensitivities to luminance it might help like zoom in on eyes potentially like that could be example of something or or um prior about like where you expect light to be coming from in general and this like helps to make things tractable but then let's say you're talking about some of like know good oldfashioned evolutionary psychology like at The Other Extreme like a you know like a cheater detection module well how the heck would you do that right yeah yeah yeah so it's like how um likely is it that you could basically in this learning

curriculum have these lessons occur and have this knowledge we acquired uh early enough and the simpler it is the more likely it could be there early um that there's um a meaningful degree of canalization that is actually part of natural selection right right right in general the uh claim I've been making is that uh in order to pull this off um a lot of it was indirect even for a lot of the the core knowledge people will appeal to um that you basically just needed to create a tractable learning curriculum to get there and what natural selection was primarily doing was sculpting learning curricula using like uh the the the body and its positioning as a large source of Leverage on this to help um reliably acquire the knowledge you need but it's still though acquired but but and how much the adap is able to do that would be depend would be um the the the simpler it is the more likely you could meaningfully make that a canalization pathway yeah yeah that that's awesome that's that's a great way of looking at it I it's funny my undergrad was in linguistics um and Linguistics was the was the hot bed for all of this stuff you know I mean chonky really um chonky and chomsky's Darwin's Darwin problem I mean this is like what he was talking about in the 90s and he had to sort of pair down what Universal grammar was because the big question was okay you can account for you know perhaps principles and parameters can account for the complexity and creativity of language learning but how the hell are these hundreds of parameters going to be encoded in the genome like in like the versions of like CH skin thinking where he's going minimalist talking about something like yeah recursion yeah so then if it's just like something like merge it's like well sure that seems very plausible right right right right even quickly in evolution like the ways it's been suggested you know and part of like what correspond to like these like phase transitions we see where all of a sudden and like new orders are coming on pretty compressed intervals yeah but but yeah but if you're going to like you know the other limit like things like I guess like a I don't know Pinker and like yes so then it's like uh oh that I'm not sure biology can do that well I think he made I think he made that concession um Chomsky at least I'm not sure about Steven Pinker but Chomsky certainly does and as you say has a minimalist program with merge um but yeah I think a law linguist might argue that he's uh he's kind of abandoned that native this project um but anyway we need not go in there I'll try and get him on the Pod actually um sticking on sort of evolution my background is not an evolutionary biology or psychology so I'm I'm super Keen to learn um again just to take a quote of yours um you say much of the functional significance of specific Connections in complex neural networks maybe inherently difficult to predict due to the sensitivity of brackets chaotic self organizing systems to initial conditions so when I hear that I think sure like it certainly makes sense in terms of initial

volatility probably having more of an effect on self-organization than once you have a kind of robust phenotype but why therefore is there a convergence within a species a phenotype to kind of a homogeneity if there is this kind of vulnerability to um to chaos it seems like a good amount of it I mean that that a good you know to get so you're having a system which has um that's massively recurrent in terms of like it's like in it in the system as it develops it feeds back on itself and then that influences Fe back on itself and that influences and so there's this potential for amplifications to take you in a lot of different directions but natural selection doesn't want you to go in any given Direction it wants that homogeneity it wants you to be like well fit to be you know uh have high Fitness he wants you to be fit to your uh the the niches you're likely to be encountering and to be able to you know survive and proliferate your forms in those circumstances and so you know you don't want be just like open you want that homogeneity but the process because it feeds back on itself will be um it should have this potential for uh astronomical bifurcation so how do you like regain it in you know there seems like and so know you could just say like you know well that's what the inductive biases are and so you know let there be inducted biases and it was good right but but well which ones and how and so I wonder the extent to which like a choke points for um coordination among subelements of the system such as for instance like any kind of place you could identify something like a bow TI Motif any place where you can get something like a large sale convergence Divergence which in general you would expect because you need small worldness for the sake of just efficiently getting a system to be integrated or to have a system be integrated all that you know you want to be able to have the different components function together but if everything's wired if everything's connected to everything that's both chaos and there's there's no uh it's very inefficient so the idea of like these places where things uh uh from Broad sense will pass through then providing basically an opportunity for um enslavement system so um so some of this could just be uh Network motifs like you look at um for instance um convergence onto like Hub structures that have high centrality but also things like if you take the value signals like you know things like uh dopamine or serotonin and you um have those also have a um the conditions of their release correspond to organism scale events then that could help basically um give coordination to the overall system in reliable ways for this task of uh helping the organism make it through the world helping it to helping the system to function so it's like these these these codependencies and from and choke points the these um places where you'll get um uh functional convergence I I I suspect that's like um important for keeping that Divergence from happening too much this uh uh

integration into organismic significance of with respect to cybernetic control of the agent pursuing its goals and and if that has perched if that basically by virtue of whether we're talking about like value signals that would then you know shape connections um or or or the the structure of the architecture itself if that is positioned such that the whole system can be so enslaved that that should help to like even though that should basically help these bifurcations in a way be uh not bugs but features and that it's sort of like they're sources of like solenoidal flow and expirations and like uh you know engaging like local pockets of like high temperature search but then like you rein it back in for staying on top of the action perception cycle and so to kind of come back to like Consciousness know for me um it's basically iterative State estimation with s with the goal being basically I want to know a Rough and Ready estimate of the relationship between the system and the world generated quickly enough that I can meaningfully inform and be informed by action perception Cycles on the time scale of their formation and so I can Nuance that engagement provide some extra skillful control and surfing uncertainty so it's like um and so like if the primary like task of the system is this inactive coupling I feel like that helps to provide that um uh that's helps to like tame and harness chaos rather having the thing sort of uh bifurcate in in you know all the like hyper astronomical number of Dimensions it could go cool yeah yeah yeah there's a lot there I mean yeah let's uh let's do the Consciousness thing um I was going to jump psychedelics but like why don't we start even more fundamental Consciousness uh anyone who's watched the first five episodes or four episodes knows that I love Consciousness uh it was kind of my for into all of this so what you've given me there is a a function of Consciousness it's guiding action perception Loops it's it's it's kind of a likelihood mapping State estimation okay but as you can imagine the follow-up question is going to be why does it come with its phenomenal characteristics um um is there something that's intrinsic is there something that's adapt to those phenomenal characteristics which facilitate improve augment the function that you've just laid out I think you know people could still you know like you know uh yeah hold there's why can't these functions you know all take place in the dark why must they like feel like anything at all I I I don't think that that concern is removed I think it loses some of its Force if the nature of the uh data structure is considered as being like specifically the generation of a visual spatial and somat spatial field of experience um Precision weighted or having like with biases rather like different areas of salience and emphasis based on organismic need and like if you're basically taking like basically the the the the thing you are estimating is the sensorium itself it sort of like well it feels like something because like emphasis on the feels like like what the

data structure is a specifically an inversion of a generative model over the sensorium of embodied agent so that's actually what the process is specifically doing now you can still say well why couldn't you do the state estimation of the system in the world in the dark exactly you know I think it's that objection though so the claim is that objection holds more for theories of Consciousness that forget we have bodies so you could still Levy it but I think it's like slightly addressed by making the primary explanandum the uh appearance of a world to a lived body if that is the primary explanatum and you're specifically saying how might Dynam DCS um compute that then I think it it takes that on a little bit more the why should there be something at feels like because what you're specifically trying to explain is the fields itself right it sounds to me quite convergent with Mark SS's sort of AFF effect driven account of qualia this being that sort of these fundamental effective states are um are sort of guides in a way to um allostatic Behavior so whether that's the resolution of uncertainty um Information Gain um testing model parameters or actually just minimizing variational free energy see that for me was one of the first functional explanations of Consciousness that really made me think oh that's that's kind of viable because I can't conceive of a feeling without it being felt um and so what I've been looking for is these kind of um it's almost like a synthetic a priori of Consciousness it's almost like a transcendental argument for Consciousness does anything like Circle back on itself so as to make Consciousness a prerequisite for the function and I think where you and Mark are converging here is quite promising which is that now someone like David oh what someone like 1995 David Chalmers might go well you could just have like an offline signal that says things are uncertain and let's resolve that uncertain granted but then I take quite seriously this uh this older phenomenological huan idea that like we're being to take a stance on our own existence and that's got to be felt to some degree um so I'm wondering what you think of of of that Spiel and perhaps where there's a points of convergence with with Marx homes's theories I'd say um very strong place of convergence but also Divergence okay the convergence is like you know he's um he he's bringing in life mind continuity he's bringing in like the necessary interestedness of the system and also like the the primary nature of of affect and of saying like and like you're saying like this synthetic op like like what do we need to even coherently conceive of this thing where like what's foundational like if you were to doubt it the whole like game Falls apart exactly yeah so unless you have like unless we get the feels in there we're not getting anywhere and unless we get like the um you know and so the the I guess the thing I would suggest uh as as a point of Divergence uh would be I wonder if the specific systems so so I guess like for for me um I don't necessarily take unconscious

feelings to be an oxymoron but um you know so we can use words differently but sometimes I'll I'll think of like one way of thinking of like emotions and feelings almost like a very abstract action perception cycle where it's like the emotions are the global State changes I don't I don't think it I think by the time you get to that level abstraction the difference between action perception falls apart but in a way though like the the the the change in organismic mode to be adaptive for a particular class of situations whatever that is that seems to be roughly something like a motion would correspond to this and where the organismic mode could extend beyond the organism in terms of the emotion just like contained within it but the feeling part potentially being some sort of um uh inference on top of that process right some sort of TR tracking of that now we can use words in any way but like I would personally um so that's that's a that's the way I think of it and the way I describe think of in that entropy paper and then I wonder where we whether we can have both unconscious and conscious feelings and emotions yeah like level where like you know we go back down to the individual cell and the cell is like changing into like a spor related mode and maybe that's like it becoming afraid or something or it like extends a pillis to like exchange a chromosome someone else you know whether that chromosome a a chromatid and that's like maybe like a lust mode or like it releases you know exotoxins or that's like a like like a aggression mode or something like a rage mode like and so like I I like the idea of like you know kind like the work of like damasio who like Mark like engaged with in in beautiful ways like you know the you know I I I love this idea I think it's essential for understanding like the actual cybernetic significance of feelings and emotions to actually be coherent about them to actually talk about life mind continuity and to recognize this is an older story than even nervous systems in terms of like the the the um the harmonics or the the igen modes of the org of the organism to be adaptive having those to be uh uh to give them an overall coherent shape but uh for the specific systems that Mark talks about um I wonder whether they uh I guess uh have what it takes to do the kind of like representation game whether um they're um potentially doing their functioning um so so emphasize like the perac Dual gray and and as you know being this um integration around organism of need and this sort of like the way that um um aexs in order for them to in order for these igen mode shifts to be coherent so that you're you're not like eating and doing Care at the same time you know like that would be a bad idea a lot of circumstances like you know you don't want like you know mothers to eat their pups you know you don't want like this um picking and the I think I don't if he says color coding but this like categorical breakdown yes I'm totally with him functionally but it still seemed like those could happen in

the dark right and so the thing I'm wondering is in order for them to be given something like phenomenality that we experience of like the conscious feelings I'm wondering whether that requires something like um basically a re-representation into cortical circuits and so he'll um and maybe even within uh within somatospatial and Visuospatial modalities so I make a uh someone considered would consider to be a bizarre claim that those might actually be all we have and all other modalities are basically uh kind of synesthetically cross mapped into those to be given spatial and and temporal spatial coherence so that they actually can be modeled and you can do this like State estimation so the idea of like your inter receptive States might actually be to be experienced have to be spatialized as a kind of like texture or as a kind of like like I I feel I I have to I have to temporize and spatialize that information and maybe part of the way you do it is to unfold them into another modality as part of the inference so um but uh yes so I guess the difference with is I'm wondering whether his account is like like like right on on the functional level but um the systems though that would actually but still though the systems he's emphasizing would be necessary but not sufficient conditions in order to obtain sufficiency they need to basically um enslave cortical circuits and that those might actually be the constitutive causal um the constitutive realizers of experience potentially by virtue of the fact that like again we I guess we get to like things like small worldness um networks that are basically collocated enough that they can Converge on attractors and a cheap coherence on the right time scale the idea of like if you're like way far so the idea is like if you're way far away from like let's say like some sort of rhythmic attractor over um uh uh is established that's um calculating something like your sensorium or like entailing something like the inversion of your sensorium on the time scales at which that sensorium is inverted um there might be an in and an out in terms of like who is able to get their message in a time and it's possible that the systems he's identifying could be um on the outside of it in terms of the time skills at which that cause of closure occurs while still being completely necessary for their functioning I I just don't think we know and so the evidence is you know he would then push back and then say like you know oh well um the uh let's see uh well no you know I I leion the insula And the emotions are still there so there's yourway and there's it's like yes um but uh I guess in a way you you might say that that that somewhat supports the the that quasi synesthetic account I just said but um I basically I I find the lines of evidence that are cited to be um not but none of them are quite convincing to me so like so for instance like um uh like hydraulic children like yeah you know they're still showing these rich emotive explains but the issue is though uh we don't know yeah like it could be unconscious emotional you we can't ask them we don't

know to like from the inside or then you might point to like um like decorit rodents showing Like You Know Rich like you know adaptive emotional uh responses and and sophisticated behaviors but once again we don't know um or you might like drop an electrode and then stimulate these subcortical areas and then you get experience and the person reports to them but then the problem is well the cortex is intact then and so how do you know if when you Dro that electrod like you stimulate the hypothalamus and then you got some sort of Rage response or a lust response was it that that was the realizer or is it as I'm suggesting that that was basically um part of um you know it it was uh causally important for the whole thing to work but not necessarily constitutive of the process that um computes or realizes the the quails so this is cool this is cool yeah I feel like there are two things going on and I'd like to sort of see if we can converge them so one is a kind of structural functional description which which is augmenting um or just providing color to Mark's view which is that we need these kind of uh recursive Loops back to the cortex it needs to enslave the you know the these subcortical structures need to enslave the cortex now someone listening might go but that's just a further functional structural anatomical explanation on the other hand you invoke kind of Proto back to the homunculi these kind of Proto conscious agents we got like whether that's a cell a cluster of cells and so on and you've spoken as well about how you know one of your your account of Consciousness you know can be cast as a convergence of active inference and integrated information Theory um in you know integrated information the go some well it got some bad press earlier this year for being pseudo science um on for those listening just on audio I am doing speech marks um but one of the claims was that it was pandering in some sense to pans psychism and perhaps an uh aute listener might be thinking oh Adam is maybe pandering attached to panm with his idea of lustful uh neurons where do your sympathies lie if we take this sort of broader philosophical position um where do your sympathies lie with pan psychism and let's just say more kind of uh I think they're called conscious realist positions such as that of Donald Hoffman example I'll regret saying this but I think you could call me a pan psychist of swords still and that like um other sense of um anything that's somehow managed to thing and coming back to the good regulator theorem on some level had to entail some sort of modeling information whether it was this the system and it's interaction with the environment was the model or is a is a model that within it has models um but has to have um some sort of tracking or predictive information uh that allows it to basically reverse resist the most uh temporarily uh res um uh defy the second law while serving in everything it does but basically that you know you getting all mixed up and becoming maximally probable

maximum ENT R distribution like how did you do that you somehow did something to do that yeah sort of intelligence some sort of thing like a mly some sort of predictive information had to have been there for you to pull off something a little bit like Maxwell demon demon esque you know to do that and so and that since like I'd say I'm a pan psychist and that like anytime we're pointing to things anytime there's attractors um you might want to say there's something mind-like going on something a little bit intelligent um but the um you might even want to say like some psychological kinds you know like like we're talking about with like talking about like the individual cells you know Rage or lust you know um like even those might be like good ways of thinking about them in terms of describing their cybernetic significance like it's a you know coherent Global change of state with a certain semantics that's linked to modeling and so if you think of you know mind as modeling and then as modeling with interest and EV Value in way you know you might call that sentience then but I guess I want to reserve this word Consciousness for the it feels like something and I want to Res and and there's there's something that you know it feels like from the inside to be that system and you so all these systems would have a kind of uh dual aspect part to them because they would entail an information geometry and so you can say it's like well you know maybe everything that exists it's just like that's the way of describing the dual aspect nature of them um I think you can make a case for that but I still want though like um basically um okay so in in so integr World modeling Theory the thing I attempted was to um take the major theories um specifically focusing on integrated information Theory and the global workspace Theory and then basically use you could say probability Theory or machine learning or this the free energy principle this idea of the system modeling its environment probabilistically as now providing the semantics and removing the question begging from these theories in terms of or Su of it of why there should be something that it's like to be a complex of integrated information or to be the um to have Fame in the brain and to be within a global workspace yeah so why well if what that global workspace is doing is it's something like a very efficient architecture um for all for like some you know autoencoding hierarchy if it's like an arena for basian model selection allowing for this iterative state estimation well then we know we're the questions are somewhat less begged or if like integr information theory is describing you know um uh the nature of networks that can do interesting forms of compute it's like okay um you know necessary conditions you know likely you know in information the they they have these these axioms they're that they look at that apply purportedly to all conscious experiences and then they try to identify these Mech they postulate mechanisms that could realize them and then the

idea is like if you start out going from these axioms of character experience you then create a precise and specific mapping to particular mechanisms that would be that would allow for their realization now you can say this system's conscious and that one isn't and maybe even this system is conscious in that way and not that way and you know I think it's it's really promising but but the issue um I take is um uh basically um in a way like like losing the explanandum of uh our sensorium our our sensory our sense world you know our embodied experience and so if what that uh if what a complex of integrated information is doing is it's basically describing a system set up for a certain type of Dynamics within it that could enable a certain type of um message passing and um and and inference and you know belief propagation inference and specifically with the thing being entailed by such a system would be the you know efficient inversion of a sensorium over uh inversion of a generative model over the sensorium of embodied by an agent now we're getting somewhere so for so the idea is saying okay um uh so I guess I'm sympathetic to both sides and viewing integrated information Theory as um having some elements which are unsatisfactory the but you know I don't think it was that particularly I don't think what happened was particularly helpful in terms of Pseudo science and you know uh for a variety of reasons but so so yeah so the basic approach is this what if you um take the existing theories the major ones cross reference their claims look for points of overlap and try to use basically degree of convergence and the forms of convergence and the coherence of those stories to adjudicate among different claims and so that was the basic method and so um you could think of for instance um you know integrated information Theory will talk about um potentially there's a posterior Hot Zone as the substrate of Consciousness and you know for instance I say that sounds about right that for instance like you know if basically the posterior if you take these different sensory hierarchies all coming together into one multi hierarchy and if you can get um basically across that system its ability to uh loop back on itself in a very powerful way in a very um you know both integrated but also differentiated way then that could be that then you could think of that is basically um that that could entail the inversion of of a generative model over the sensorial of an embodied body and an agent maybe that's what's going on you could also potentially think of though this complex of integrated information that has both uh integration that where the network properties are set up to facilitate both integration and differentiation that's actually has a lot of correspondences to workspaces which mountains integrated segregated functioning and so the idea as like that poster hasum could be thought of as a fairly Global workspace you know and um with details in details of exactly the kind of workspace it is and you know with um with me having some specific proposals so then like

for instance uh okay so the workspace Theory they might come in with the claim like no no you need the front of the brain and so the question is well what are we talking about exactly are we talking about um phenomenal Consciousness like the actual generation of your experiences from moment to moment or are we talking about your knowledge your of experience your your knowledge of your experience your awareness of that you're ability to report on them and contextualize it in different ways well then you're probably right the B is probably right that you you need to get the frontal lobe in the mix to do that although at that point though by the time you're you're talking about things like that you might even need more than the frontal Lo we actually might be in like an ex like a extended mind account in terms of like what actually goes into to be self-reflective like you know we like we we you know use like the environment to like offload memory like the way like where so you know the workspace the global workspace or play Space might not even be skull bound you know but still the so yeah so the approach I took was basically to say that um the different theories were bringing important contributions that all these brilliant people that these are all brilliant people who've given a lot of thought and there have something important to say then the question is are they may be using words differently so like in that debate earlier between like integrated information Theory the global workspace Theory you had this recent like adversarial collaboration say but it's not clear they were using the word Consciousness in the same way like integrated information Theory is talking about experience like and phenomenality Global workspace Theory sometimes it's like illusionists sometimes you know they're they're more it's like more like a higher order thought type Theory and and they're oftentimes not necessarily but you know it's it's usually they're talking about you know um you say like conscious access as opposed to phenomenal Consciousness and so so then the idea is that if you're you try to be more like like what are the precise claims that people are making and you dig into them a lot of the conflict might just be be people not communicating enough and saying like oh I meant this by that term and you meant that oh and instead there being just like one way we use words it might be that like the different ways we were using words had different trade-offs and maybe like to come together we have to like add qualifiers so instead of like we try to say this one word means this one thing maybe we use like um we need to use like two or three words to specify which World we're in and what we're talking what what our reference are and then the claim I make is that once you do that um a good amount of conflict actually is not as um much as it seems like it needs to be like you you're actually significantly like these different theories are actually to important degree and a useful degree talking about different things right right yeah

well you know maybe there's a psychological account of the fact that scientists like at you know they they might pretend that they believe in collaborative um work but there is there's a there's a tendency or an in towards adversarial boundaries which are often incredibly constraining so no it's it's a good point and I think it's a good point as well about how much language really shapes again I I don't mean this in some sa wolf deterministic way but shapes the way that we view these um these models the way we view these debates I mean I think the most I think for me comes back to to what you're talking about like right at the beginning that I could look at a rock and say well there's something kind of going on there it's tracking some you know it's doing some kind of variational Basi and inference one of you know I think the two most important words for someone coming into the free energy principle depends on the definition of the free free energy principle you like but my two favorite are looks like so anything that has a Marco blanket and a uh you know a steady state tractor non-equilibrium steady stere tractor will look like it's conducting basy and inference and that look the the the question of what it is doing the process Theory really gets opened up once you go oh the free principle is just you know it might well just be a descriptive tool and then we talk about active inference predictive processing variational message passing but I think this this looks like notion is often um forgotten and it's important to keep it in the back of our mind that yes there is something quite miraculous about the coherence of a stone um but I guess every theory has its Miracles um and the fre principle is not it's it's not devoid of its axioms do you think that's a kind of reasonable thing to say I think so and it's like yeah in general we're trying to do is like you know I guess would uh minimize the number of like you know DX moach as we need to call and the number of Miracles and like but you we're need to like like you mentioned like synthetic opary categories like these things you need some you need something to build with you know or also in terms of like the what we're talking about with like uh evolutionary priors and like inductive B like you need to start somewhere you you need something must be granted you know but but like but let's try to like do the most we can with the least and so um and but but still those things that you um are granted you know in a way like there's a sense in which like you they might be like they're the preconditions for proceeding it all those are the Miracles we need to to go on and so there's like we end up like uh it's like for the case of like yeah like a rock and you just say like you know it looks yeah I had a section I put into like the like original and and and and left it in there maybe unfortunately like like the like the iwt paper and I I I felt crazy writing it but you know I think like if I like emphasize this like looks as if like it feels less crazy and then saying the rock is engaging in basing inference to exist

you know but I guess you know there's a you I guess you could say there's a sense in which like that's how it is at that point I guess you're almost in like a quasi tautology and that like like let's okay so here's like the face based description of the Rock and then it uh I guess you say like if you keep looking at it um well I guess all the rock though it might be we need something like it might not work for the rock but it might need the like you need something where like that density of the thing like like that you can point to it and like there's a combination like the solenoidal flow and gradient flow where the thing is like picking itself back together like a drop of oil yeah like a drop of oil that like like re that grabs itself and then yeah yeah that yeah so once you do that is that kind of like I guess that's Carl's like 2019 paper then it's like a sense it's like you can say like well that actually is Basi inference yes that there's like where you expect to find it there's a thing like as a probability distribution and then if you just look at its action it's doing basing inference and then like well is that a tautology maybe but um uh it's a damn interesting one tautology and on yeah yeah yeah no it's uh I remember hearing Carl first on Lex freedman's podcast talking about a drop of oil kind of my first proper exposure to free energy and I just like what I just couldn't think I I I didn't believe that drops of oil or rocks or molecules could be so interesting and and you're right you know um science I don't know who said it but science is about making the familiar unfamiliar um and we're certainly making rocks and drops of oil pretty unfamiliar when we're talking about then conducting variational Bas and inference there's someone who might be interesting to talk to he wrote a paper on like the world is neural network and so like I think he would um potentially get to this issue of like um is there just like a way in which like um you could just make like an like the ontological claim like the the rock is inag like he like he would do I think a a good job of like making a case for that potential okay cool I'll check out the paper I mean yeah I mean for me this the question that I literally started this podcast with still stands which is if we take the drop of oil it's it's it's kind of yes can be seen as calculating a posterior but what does that mean like what does that mean like what is doing the calculation and what's it doing a like it what so the the P of itself given its environment is just really high I mean once you sort of break that down it just becomes it it seems to me to just return back to a descriptive account rather than a uh process theory of what the what the drop of oil is actually doing so you know I think where a lot of active influence confusion comes from is that we can't um we can't we we we struggle conceptually to get away from aties in ego model when we're thinking about how we are and how the world is so it it's tempting perhaps to think that there's a little homunculus within the drop of oil grabbing itself together and maximizing model evidence for

itself but even I would not say that no no neither that was by the way I'm not saying that either but it's um it might be tempting for some let's um let's get to psychedelics because you're at John's Hopkins which is the kind of place to be in terms of psychedelic research I think a useful place to start not only for myself for our audience is people might have a sense of the history of psychedelics in science um and the long history and the taboo and the fact that now these experimental studies are taking place and the taboo is lifting perhaps you could kind of give a uh brief history if possible of the history of psychedelics in science um you don't have to go back to shamanic cultures but if you feel so inclined I I Shan stop you I mean that actually is a um I'm working on a paper right now of like the Deep history of psychedelics like um going both back to shamanic cultures but before in terms of the um evolution of the um signaling systems that they appear to primarily act upon uh in particular the um 5 ht2a seron receptors and then is a Agonist all that receptor that particular receptor class yeah and um so there there's a a paper I just recently uh revised and resubmitted um called uh um Albus or alter beliefs under psychedelics which um basically provide uh is it's providing I guess you might say heterodox account of psychedelics um within Act of inference I read the preprint so uh I'm I'm familiar so so that's um so I guess we'll get to that in a bit but this so uh that's like where the focus has been but the current paper would be like um why is it that we have psychedelic experiences and like like what is it is there is there like so you you might say um maybe in a deflationary sense you could say like like pink her would talk about like uh music is uh cognitive cheesecake where it's like you know if you try to look for an Adaptive story about music you could try but it's just like no there isn't really one that's like that meaningful it's sort of like you can the Adaptive story of like cheesecake is like it's a super stimulus that's created because of this this and the other but like there is no evolutionary count of Cheesecake it's a late Comer to the scene and so it's a deflationary count of like music would be like there's no specific music adaptations it's just like a super stimulus we create for ourselves you know uh because of other things that we need and other aspects of our cognition and so with psychedelics you could say like you know these drugs just happen to be like you know a super stimulus that's like created um you know maybe by plants to defend themselves against predators in terms of you know you don't want to quite um you know uh you know you want to um either repel and so you know maybe like you know breaking the good regulation like like like making the thing do anomalous infer inference could be you know a good way of like um repelling you your your Predators um or you might want to potentially entice in terms of you know there's like some symbiosis that exist in terms of things like that was it um reindeer

um I don't know if this has been established but like you know um possibly like by consuming things like um is it like is it lyen I forget what it is but like fungi like it might like help them to be more like exploratory in certain parts of their like seasonal cycle and so then there's like a symbiotic relationship um and you're actually enticing the organism to come in but the um yes so actually that's that's a very Live question of like the Deep history of psychedelics in the place I'm going to I'm trying to go in this upcoming paper is to argue that um you know so the it seems like the receptor system itself um might have evolved I'm claiming um actually in a way so so um so the count of psychedelics in the frery principle I think that's most wellknown would be called the Reus model yeah relax beliefs under psychedelics and the idea being that psychedelics are relaxing your basium priors and so with your prior expectations relaxed you are now free to think different thoughts and be updated by the world experienc it more profoundly and be changed by more more profoundly and um so I've been um trying to uh Nuance that account and saying that like that might be um part of what happens as part of like very high degrees of 52a agonism and that you do get something like that but that might not be the cybernic significance of the 582a system as a control parameter um and that in in many ways it might actually be exactly the opposite and so they first evolved with a uh Gene duplication event around the Cambrian explosion around with the Advent of JW fishes and so one of the um speculations I make is actually and this might be um Predator prey arms races where someone can basically if you can specifically bite then this now like adds fuel to um a certain fire of like you know I can now go up and and precisely swim and bite and kill and now you have an interest in being able to have that strong prediction that you uh use to basically uh enslave your in action and Achieve that goal um and and the stronger you can make that prediction the the the more you know potentially within limits the the more reliably you'll Converge on that goal as opposed to doing something else but then you might also have an interest on the side of prey of uh not being the person bitten sure and so this idea of like uh turbocharging the organism um doubling down on the priors potentially as a way of allowing for a specific goal Pursuit that might have been the original story but where does this like Reus account come in uh one of the speculations I'm making um along it's it's a a paper I'm writing in collaboration with Sue Carter who uh studies oxytocin mechanisms um and there's a follow-up paper I'm writing with um before the other one was published with um Matt Johnson but um is that actually um orgasm and birthing might have been the original psychedelic experiences and that actually so you're basically relaxing the priors about the bounds of yourself and having greatly elevated plasticity and capacity for updating why would you want

something like this those would be two cases where you'd want it whether it's bringing basically two basically bringing either a a letting allowing a a mother to care for the child long enough to uh help it to get past these early milestones and also allowing potentially um uh mating pairs to stay together long enough to help these hungry children to acceleration okay that's that's G into that then so what would the the kind so there's some coupling going on here in terms of this kind of turbocharging it's Psych in a sense it's kind of psychologically adaptive to have relaxed belief over yourself over kind of yourself model maybe even your autobiographical self model um so as to facilitate um the Advent of another person into your life right so like people just qualitatively talk about how they now feel like this child is their purpose they're kind of secondary to this child in their life so what's the kind of what's the causal par for and what's the chronology in this is it is it a genetic adaptation that um occurs kind of simultaneously with like modern child birth is or whatever that looks like or is it um does it predate it and then get sort of exapted by um birthing star yeah birthing or does it come later that like what what I don't know actually how you would examine that but what are you kind of claiming in that regard so the specific um the specific mechanistic Pathways that were canaly in which ways uh we got no idea sure so forance like uh um you know there's there all sorts of things that could contribute like for instance um uh there seems to be for instance like the the 2A receptor um you know in general it's it's it's actually fairly insensitive relative to something like the 1A receptor but if you take it and expose it to um like uh reduced like blood acid like rather reduced blood pH or elevated acidity through something like uh hyperventilation like you know some or things like um carbon dioxide or lactic acid accumulating then it gets potentiated and so like for instance like you know during uh sexual activity during um uh uh birth there will be a lot of heavy breathing um uh over an extended period and you know is that actually like an accumulation of carbon dioxide and actually that's part of a mechanism of potentiating these receptors maybe we have no idea could it involve something like endogenous DMT maybe we have no idea could involve as Sue is arguing um uh oxytocin vas suppress in different combinations quite possibly um but we we're just beginning that now in terms of the specific Pathways but the in terms of when this might have occurred and some more things that could have been important I'm wondering if there were like a couple waves of strong selection one probably with like the Advent of um Mamon uh reproduction in terms of like and Mamon care so so the um you know once you're now having to going to earlier like don't eat your child care for it you know once you're getting to and and youve seen a lot of rodent species it's actually you know a little bit uh tenuous that process

right like like the run to the litter often gets eaten you know you're not that far from like the atavistic state of kind of like of um so to get this interest like U was uh I think Pat churchin will actually talk about like an extension of like self grooming to down so for instance you can maybe think of like actually like relaxing body maps and then making them plastic for a Time on the occasion of birthing or like for instance like during nursing and then like basically if you then include um the pup into the body your Offspring your body Maps like that could get in count like that off the ground um or then once like for instance the um to the degree you get um parental investment you know but this would then be like a a mamalian story and any of those species there might have been strong selection on like these like protos psychedelic Pathways and then you know for and then for humans you know it would be like all the things coming together in terms of um having uh you know the these big brains extremely vulnerable hungry Offspring which you have to care for for a long time oftentimes also needing you know more than one because part of it also of the Care is providing um you know protection within the social hierarchy teaching them things you know two heads being better than one and so having the you know the bonding of the parents for the child and the parents to each other like now the selection pressure might be really strong what was selected you know something involving oxytocin essentially in connection with DMT with other we got no idea but wow then I guess the one more thing that could happen in um not just in humans but maybe particularly in humans is our um our ability to like like was it Jonathan talks about like the hive mind switch like our ability to actually like have Collective effervescence and to come into like a group a sense of group identity for the sake of basically uh um uh doing things according to Fashion including beating out other groups and so there seems to have been strong selection on that in the human case also and so the claim would be that these like basically pressures for inter subjective uh coupling being particularly strong in humans might have been a source of why for instance um uh psychedelic experiences uh we're probably not the only species to have them um but the idea being that um at least the reasons that we know that we are sensitive to them some of that could be because of that history of selection because of those challenges of of of bonding between mating pairs between parents and children and potentially between uh tribe yeah that that is really really um it brings two things to mind um one is I don't know whether you've ever heard of the obral Dilemma um which is basically this idea that when humans went from being quadrupedal to bipedal the hips of the mother narrowed and what was happening is there was a lot of uh stillborn deaths a lot of child birth deaths because the hips were narrowing but the size of the baby's head stayed the same so a lot

of children. In human child birth a lot of mothers were dying in child birth and so there was an evolutionary tradeoff between early birth and bipedality and so what actually happens is that children start to get born younger pregnancy is reduced in time to permit the narrowness of the hips that allows for bipedality which in turn sets the stage for caregiving and the fact that a child can't just pop out of the womb like a gazelle and go running so that's the one thing that comes to mind I'm gonna include that in the next paper well give me a shout out and then the second and then the second thing that makes me think of is maybe this care domain this fundamental need to expand body models self models could be this kind of Consciousness synthetic a priority we were looking at maybe there's an Adaptive pressure to feel this kind of Selfless Love Agape whatever you want to call it towards your Offspring so as you know for evolutionary reasons so that you are a caring uh altruistic caregiver that's I think potentially very important for like so this in the evolution of cogn modernity there seems to be a kind of like chicken egg story or site dilemma where um you know so like chsky like you know you know talks about language for the service of thought but like so I guess that he somewhat sidesteps his dilemma by saying like language is just useful for structuring thought right from the GetGo um but like let's say like you aren't quite thinking that way and like language is about communication like At first like you are you think you need that to get like language use off the ground well then the question is like um but if you need language use to like who are you communicating to and so like something that could basically um uh scaffold um so so in terms of like the the um elaboration of uh forms of Consciousness and maybe like the even potentially like the Precision and the and the uh I don't if You' say richness but like like even consciousness like phenomenal Consciousness in its qualities its ability to you know uh represents like precise things like it it seems like in kind of it seems like if there's like a use to being able to feel your feelings in a way where you can reflect on them in a way where they can meaningfully uh uh impact your behavior um if that is helping to um basically provide some of the selection uh for certain I don't just thought works but for certain complexifications of Consciousness that that could potentially help other places where you might have chicken egg problems like as a source of pre-adaptation like I guess is what I'm wondering yeah exactly I think that that's exactly the line kind of shooting down yeah I mean this is yeah wonderful wonderful ideas I guess another question that comes to mind is when people think about psychedelics they probably don't think about relaxed basium priors they think about tripping for on of a better word they think about hallucinations Visions auditory um hallucinations as well so in terms of the sort of adaptive self-organizational um yeah uh

functional utility of psychedelics that will make sense where on Earth do uh the sorts of uh Magic dwarves or uh recurrent patterns or whatever Terrence McKenna was talking about what what's going on there and and how does that complexify this story so um so before we get to machine elves so part of the heterodox psychological view uh uh rather the heterodox active inference on psychedelics um would be that um potentially those types of phenomena as hallucinations or some forms of like you might say psych psychotic cognition or like the the um you know the entertaining of of a of a thought that you would not normally under entertain you might think to be crazy under other circumstances but that is there live in your mind that you can maybe think of in some ways it seems to have like almost the shape of a schizophrenic delusion or like a Proto psychotic delusion so um you could potentially argue for these as a relaxation of highlevel constraining beliefs allowing those to be free form but you could also think of those as particularly strong priors also that you're imposing onto the world and not rather you're now saying I'm gonna I think this is there this is my expectation and I am not going to let that be contract contradicted by sense data and so that's so there's an interpretation also in terms of a strengthening of belief of those phenomena you mentioned and one of the points I'm trying to make in the paper is I don't think we know how to think about this and so it's it's probably some complex like ad mixture of relax and strength and beliefs both directly and indirectly in different combinations um where I'm I guess trying to say the like um rather than ever saying like psychedelics do X I want to replace that with um this psychedelic uh this particular compound at you know roughly at you know this range of doses does this to this kind this kind of person with this kinds of set and settings and the idea of saying that like but the idea of like taking inspiration from Reus is like relax Bel being an essential feature in some ways and looking to adjusting of your priors and your mind is this like Leaf ecosystem as a fruitful way of like contextualizing psychedelic functions but then saying um moving beyond relaxed beliefs talking about like both strength and relax in direct and indirect ways but to come along to uh DMT elves that would be like an example of like um you know so you can say uh you know uh you could maybe think of that in a relaxed belief story and that for instance uh know elves do not exist and now I've like relaxed certain aspects of my ontology and now I've like made room for these imaginings to arrive in a vivid form like because of like um of an indirect because I relax beliefs at this high level of certain core schemas that like tell me what I should expect in the world now I've made room for psychedelic elves for instance or like you know entities you could also tell a story though that these are actually um kind of coming back to core knowledge you know things like um uh um

uh agood or agency or things like um but maybe not even core knowledge like just early lessons in the learning curriculum and on Ong going ones of just we expect agents in the world because that's the primary thing we're encountering that's our primary niche as each other and so then basically for the beliefs so like in a strengthened account you could say let's say you're strengthening your beliefs maybe that you're the you're getting a direct strengthening of these Prime these priors for agency detection are they evolutionary priors are they developmental priors we have no idea you know in my thinking they tend to be more developmental but it's you know it's a spectrum of how you know where you how how how much Evolution shaping went into them and but as as our conversation earlier um yeah also I I stumbled across uh you saying just these words which really call my eye archetypes as call priors and I thought oh that's fantastic what a what a lovely way to put it I mean okay I this is just this is really really cool um really exciting I guess guess so so just to piece together to scaffold together the chronology the causality the phenomenology let's just like try and put it all back together so if the original psychic experience let say was uh child birth or orgasm how does this end up uh is it just a is it just a coincidence that this ends up in Plants if it is it is informative that it was happening in child birth or um orgasm or is that just kind of a as I was saying like a coincidental feature of the activation of this receptor um because your you the argument that you said there which earlier which is that is a kind of defense mechanism or a kind of uh attractor toward symbiosis is quite convincing I guess we have to fold into all of this the shamanic cultures and then the kind of more modern therapeutic lens on psychedelics is there a way of coalescing those seemingly disparate strands so like for plants like there's the impact that these signaling molecules these different ones would have on the plant itself and then on the other systems that interact with them so for instance the um you have the 1A and the 2A receptors will tend to have uh different impacts on cytoskeleton Dynamics and so let's say you want to do something like a phototaxis like like that actually it seems will involve some um I'm remembering correctly I should remember this precisely but um there might even be like you can think of it like um changing the cytoskeleton Dynamics to allow for greater fluidity you can maybe think of that as like a kind of belief relaxation for Mo morphogenetic inference of the plant so it's like you can you can think of the plant as like you know you doing a phototaxis and say no I'm here now I'm there you maybe you think of that as like a kind of like um uh updating for the plant itself and and maybe that's even part very speculatively of why like you find it in something like a cactus where like that's actually kind of a harder thing because you're block here and like you know something like that but then the

other aspect would be the defense and another reason you might have it in something like a cactus is because like if you're in the middle of the desert and you're juicy you're extra tasty you know like in terms of like there is a lot of game there there are a lot of you know contenders in town for things to eat there's actually plenty everywhere but still you're you're you're a Target and you're stationary um by and large you know so then like you know the which Predators were they targeting you know like like insects um you know uh what's happening to the belief Dynamics in the insect and so you know at that level you know maybe you know it's largely a uh uh um a Rebus account of of like these 52a agonists for the insects potentially in terms of like changing uh their normal behavior I'm not sure how you would think of their but like the idea like you know someone you know nibbles on you um it might be good for them to uh go away become more exploratory and and maybe even um mate like crazy like you actually see something like this in like uh locusts um they have they're colonized by hallucinogenic fungi which seem be part of their mating cycle and is part of this like period of intense mating they undergo and so you know you know you know the question is like why not just poison them you know maybe it's just like this was like the best poisoning came up it was good enough and it just did the job it arrived by some means and just the form of poisoning is um taking the the model the organism has by which it is adaptive you relax that model but that's not necessarily a good thing for that that that that insect especially you know with his body weight you know it's eating a lot of it and so the idea of that that insect you know if it's if its models are really altered and if those models are trying to constrain all the different ways that system can bifurcate to like be more homogenous as response and adaptive ways um you don't come out necessarily ahead by doing that adaptively and so that's just the way um insects figured out or their plants figured out how to protect themselves um is and and the point is the attack uh but maybe it's not even an attack um maybe it's like you want um you know it's you it's fine if you nibble on me just don't nibble too much and then uh but just go away and then um and pre and and then if if we had this nice relationship of you don't bother me too much um you know maybe I even like you to be around maybe I even like that I facilitated your mating and I like that your genes are more prevalent in the Next Generation so yeah I guess and then coming to you know Shaman well before before you do just I'm curious about what you think about this because just to like add maybe a a technical en this I'm curious about what you think from a Rebus account in in this case you mentioned exploratory Behavior there or epistemic foraging which I thought was really an interesting illusion because in both cases you offered there there's a move towards

epistemic Behavior either whether it's mating and or you know go away in an aversive manner or go away um and basic like the reindeer example you know spread me around basically in in a kind of weird way right yeah yeah yeah whates go forth right and what that screams to me is so just in in the sense of like the actual computational modeling in pom DP schemes is an emphasis over expected utility would be sort of high Precision over your B and C schema um and and e as well your prior uh your prior overaction your habits basically and maybe what's happening here again it's a completely nent fault that I haven't given much obviously uh you know fa to is that rela you're relaxing your prior beliefs about the consequences of your action thereby making every single possible action policy equal in terms of epistemic utility which means that you then Maxim you do some sort of jany and maximization of entropy which makes the best uh in terms of just the um expected free energy formul makes the best action policy that which maximizes Information Gain and all uh test model parameters so I'm wondering whether that speaks to in terms of a kind of technical unpacking of what might you know what might be happening at a very um low level here with insects and cacti it was fit well um with like changing migration patterns I think it would fit really well with that um in terms of uh like like think of like human use cases it seems like it's a mixture like so for instance like um so like um sometimes you know specifically would be for um you know breaking out of some State you're caught in like you go to the shaman and this is part of it but but oftentimes it would be like with a very specific interest in mind like um who someone cursed my cow who is it you know or like but or or sometimes it would be like um uh uh so sometimes it seems like it is a good amount about like the epistemics in terms of about like trying to maximize learning and breaking you free from like the thing that is usually owning you from within you know like your usual way of achieving your goals keeping you stuck in a certain pattern and then it's like you need to learn something different but how are you going to learn something different if you're always doing the same thing again and again and so yeah I I think that fits for a lot of the like a lot of the healing uses that a company like like an iasa ceremony that that seems like a good description but there like other use cases like for instance like in um like um think it's Amazonian tribes like uh it's actually like specifically pragmatic and that like they do it to to get the hunting party together and like they'll um they're about to go to war with a specific other tribe but then you could also though make a case of like um you know is it that like their like their goal of what they want to do when they encounter the other tribe you want that to be presented in a very Vivid form and that imagining you want that counterfactual to be hyper precise so that becomes actual or is it um more like a matter of like

this kind of collective effervescence from earlier and you actually want a Rebus account so that before you go into battle you're basically you know all one body coordinating together fighting you know having each other's backs um or both and you want you know an early stage of something Rebus like and then maybe a later stage of something more uh involving belief strengthening so it's like sometimes you're letting go of the pragmatics and sometimes you're actually like holding on to the the um pragmatic values specifically zooming into it um it seems like there like at least in the human case there is no singular account but in terms you're saying about like things like um uh like the the case of like migration patterns in an animal being changed um that seems like that that that the relaxing the pragmatics allowing the epistemics to to more optimization for that that that could be a really important part of that I think cool amazing yeah I mean we haven't got too much time left but um I guess two more questions one totally trivial the other one not trivial at all so I'll start with the non-trivial one um I Hawkins you're you're really at the heart of the current uptick in psychedelic research are you well just to frame this I sense from an external perspective there was a great deal of excitement you know starting five six seven years ago and maybe now there's a um there's a slight conservatism that's crept in and and and saying okay the the psychedelics might not be the Panacea that we hoped it might have been seven or eight years ago again I'm not on the inside I don't know but that's just my instinct what what are your kind of uh predictions expectations level of optimism with respect to psychedelic therapy in the next 10 20 30 years um I have some uh so if we can avoid the very live possibility of uh backlash as um uh adverse outcomes happen inevitably and they will be more notable you know like with driverless cars like if you know even if unet they're saving lives you will note those cases where it just slams into you know a person and you'll just be and that'll change everything and so it doesn't take many people to have some really bad experiences um you even see like with the re like Matthew Perry like what happened with him AI and and that really is um actually really deeply responsible for everyone involved because um I don't want to say that you know like ketamine's you know uh not a mixed uh like it's powerful medicine and but uh in that case like um you you see the shape of it there's already like people are interpreting this as oh ketamine is dangerous and already this is causing people to call for more of a of a pullback but he was also on a a good dose of an opiate um probably had some drink system and was in a hot tub so you know so it's like um you know did he die of respiratory depression and then drowned you know it's uh but so things like that can happen and uh it doesn't take many of those for the Psychedelic Renaissance to really be like pulled back in ter potential like

lifechanging uh medicines interventions um oh by the way uh whenever I'm speaking it's um outside of a h Hopkins position they and they are um so speaking from the outside um they have a particular role I'd say in in the Psychedelic Renaissance of doing largely conservative work within a mainstream medical model and trying to say can we push forward that and and and how can we approach these medicines safely and responsibly and so um they tend to be I'd say more conservative in what they're doing and it's a very valuable function um and and so uh you know in general they try to be very careful I'm I'm a little less careful um so uh the um some of the things that are so some of the open questions and challenges we have is what um I guess you say business models or how can we scale psychedelic therapy so I so one of the problems with like like a psilocybin um intervention is they're amazingly expensive and so the question is like how can we get like broad access to them you need some sort of like group therapy and like a different model than the ones we currently use just to make the economics work to make this like an available one a widely available intervention um the other question is like if you remove some of the um expectancy effects you would get from like the places where these interventions were done and like um you know the you know having it done in Johns Hopkins you know like the um or the Good Friday experim you know and Har there's a certain trust and Authority which is interacting with this as part of like leading up to it that's part of like the expectation that you will be so changed and so if you then try to you know have clinics opening up everywhere do you get the same effect because you'll have you know the reports you know is some of the five most meaningful experiences in a person's life a few sessions your life has changed and to what extent can we still get outcomes like that um as we move into basically scaling and and having more institutions in more context and do so safely one of the things I'm actually very excited about is I have some uh friends in um Pharmaceuticals who are basically creating analogues of these psychedelic compounds to try to change their property slightly and so one of them is like for instance um uh shortening their duration of action well why' you want to do that well let's say like you could show up for a therapist and you can do basically MDMA therapy and be sober again in two hours m and so that would both allow you basically to do it um more flexibly do it uh and do it more times potentially and you know and with less risk from the dose because things like cytotoxicity will be a function of both the dose and the duration of effect and so if you reduce that duration of effect things like you know um metabolic strain you would reduce it so those advances I think going to be a complete GameChanger like uh the these short acting powerful compounds I think Psychiatry will not be the same once those are on the scene and just lives

will be saved and transformed fantastic so yeah and so there's a lot like things are very fraud right you know we could have something like a couple bad outcomes could cause like a a chill on the whole thing but the promise if we could just make it through this Crucible I think it's unbelievable like amazing well that's exciting and then my uh my super trivial question especially even more trivial given that response is um iasa is uh famously kind of an enigma um because it's you you may correct me on my uh chemist my biochemistry here but isn't it DMT mixed with a plant among the hundreds of thousands of plants that are found or could be found in the Amazon what's your Joe Rogan esque thoughts on how on Earth they managed to s emphasize iasa hey you have your your edible TMT but that'll be broken down in a hot minute and so you need to find basically this monomine oxidase inhibitor to keep it going but among all the different things you could have tried how did they find the right one quite yeah the shamans will tell you the plants told us that's how they did it that doesn't help um I guess you know I'm hoping I don't know maybe we should get like Joe Hinrich on the job and like basically some sort of computational mod of like the process of like trying out the permutations of like what shamans do normally uh maybe it's actually tractable in terms of like if you see just what they're doing it's like yeah they were going to find that given what they were doing in like over over 200 years you would expect this percent that they found it sure so I would expect something kind of like that or maybe the plants told them well maybe the plants told yeah maybe Adam that was that was amazing I'm kind of mind blown um that was awesome thank you so much I I mean with every guest I always feel like I learn just infinite sums of new information which makes this so much fun for me um but no I really thank you I I know you're a busy man and so I'm very grateful for you giving me two hours of your time um it was a pleasure we're going to we're going to keep digging away at these questions um but yeah for me and the insute yeah it was it was an absolute uh pleasure on my part so thank you again thank you all right that is